Structural Change and Economic Dynamics 73 (2025) 77–88

Contents lists available at ScienceDirect

Structural Change and Economic Dynamics

journal homepage: www.elsevier.com/locate/strueco

Minsky in Peru-unveiling the hidden financial fragility at a sectoral level

Samuele Bibi [a,*], Pankaj Avinash Chandorkar [b]

[a] Aalborg University Business School, Aalborg, Denmark
[b] Department for Accounting and Financial Management, Northumbria University, Newcastle, UK

ARTICLE INFO

JEL classification:
G10
G30
M21

Keywords:
Peruvian stock market
Minsky
Financial fragility
Sectoral analysis
Indebtedness growth

ABSTRACT

Since the beginning of the new millennium Peru has often been described as the Latin American economic miracle for its sustained achieved growth and stability results. However, recent works highlighted the potential hidden financial fragility that Peru might be subject to, especially at the institutional level, considering the non financial private sector. To investigate such a phenomenon in more details, we build Minskyan indicators to study a sample of 59 companies listed in the Peruvian stock market. Our analysis detects different financial fragility exposures of those firms at a sectoral level and it empirically studies the drivers of indebtedness growth at an aggregate level during the 2003–2023 period. In the context of the detected financial fragility, this paper serves as an academic ground in enquiring theoretical key questions and providing relevant highlights for the policymakers of Peru.

> "*Experience indicates that our economy oscillates between robust and fragile financial structures, and financial crises require the prior existence of a fragile financial structure.*"
>
> Minsky, 2008, p. 233–234

1. Introduction

This article aims to explore and investigate Hyman Minsky's Financial Instability Hypothesis (FIH) in the non-financial sector in Peru. Although empirical investigations of FIH in the extant literature are mature, such an investigation in the context of emerging Latin American economies is rare and especially absent in the case of Peru.

During the last decades, Peru has been multiple times portrayed as the new "Latin American miracle" (Mendoza, 2013; Vietor et al., 2015) and earned praises from the IMF for its economic expansion and performance (Ross & Peschiera, 2013; IMF, 2008; 2012; 2017; 2020). The commendable outcomes were not solely confined to economic expansion but also extended to the realm of maintaining low and stable inflation, along with steady exchange rate dynamics. Perús economic and financial stability achievements have frequently been attributed to the adoption of orthodox mainstream policies that have characterized the country, as emphasized by Bibi and Valdecantos (2023, 2024). Additionally, IMF Deputy Managing Director Mr. Kenji Okamura recently echoed this perspective (IMF, 2023). In 2020, Peru received acclaim from the IMF as one of the top-performing economies in Latin America due to its annual real GDP growth surpassing 5 % over the previous 15 years (IMF, 2020). Recently, Mr. Kenji Okamura shared the same point (IMF, 2023)

> Over the past two decades, Peru's very strong economic fundamentals, institutional policy frameworks, and prudent policy settings have underpinned robust economic growth and stability, spanning multiple electoral cycles and governments. The solid inflation-targeting regime, credible fiscal framework, and robust financial sector supervision and regulation have allowed the country to deploy a robust policy response to mitigate the socio-economic impact of the pandemic, and subsequently remove the policy stimulus, while preserving macroeconomic stability and maintaining ample access to international capital markets. Kenji Okamura, 2023

However, during the last few years, both GDP and GDP per capita stopped growing at the same previous speed. Recent US reports (U.S. Department of Commerce, 2022 a; 2022 b) argue that, if fiscal management and macroeconomic fundamentals contributed to the country's region-leading economic growth since 2002, COVID-19 in 2020 and the recent political instability in 2022 were the major causes of such a change in Peru's economic performances together with corruption and

23 October 2024

* Corresponding author.
E-mail address: samuelebibi@business.aau.dk (S. Bibi).

https://doi.org/10.1016/j.strueco.2024.12.013
Received 8 April 2024; Received in revised form 23 October 2024; Accepted 10 December 2024
Available online 11 December 2024

social conflicts.

Differently from such a narrative, Fig. 1 underlines how such a reduction in growth speed at an aggregate and at a per capita level was not only due to the COVID-19 effects of 2020, since the reduction of growth speed was already visible before the global pandemic period. These findings are in line with what reported by INEI (2020, p.45 and p.46) – National Institute of Statistics and Informatics of Peru.

Studying the case of Peru both at an aggregate and institutional level and based upon Kregel (2004) and Minsky (1986, 1992), Bibi and Valdecantos (2023, 2024) suggest that behind the economic and financial apparent stability, Peru might have been affected by a hidden process of financial fragility. According to the authors, such a financial fragility has been particularly relevant for the non-financial private sector (NFPS) but it was mainly covered by the high prices of exports during the 2003–2014 supercycle commodity boom, defined as the period in which commodity prices reached record levels in the last decades. That dynamics was largely driven by China's high demand for those products benefiting several Latin American commodity exporters including Peru. The financial fragility dynamics is related to a situation where the growth of indebtedness in an economy outpaces economic growth. In such a context several firms might find difficulties in paying off their indebtedness obligations making the overall financial system more fragile.

Because of the increased capital market liberalization policy that Peru has been pursuing during the last decades, it makes sense to gauge both the relevance of the stock market capitalization as a proportion of GPD and the dynamics of the Peruvian Stock Market since 2000. While Fig. 2 (in the Appendix) reveals that the stock market capitalization doubled its relevance from about 20 % to more than 40 % of GDP in the first two decades of the 21st century, Fig. 3 reveals the extraordinarily strong surge (and strong variations) of the Peruvian Stock Market index (BVL), even comparing that to S&P500 indexes during the same period. Furthermore, the dynamics of the BVL seems very related to the fluctuations of commodities prices in the mining sector, especially copper, which constitutes the main driver of the Peruvian economy as underlined by Bibi (2024b). The evolution of the price of copper is shown in the Appendix (Fig. 11).

That way, and building on the hint of Bibi and Valdecantos (2023, 2024) that inspired our work pointing at the potential hidden financial fragility of the Peruvian NFPS, we shed light on such a phenomenon mainly in two ways. First, at a more mesoeconomic and sectoral level, by building indicators upon a sample of 59 NFPS companies listed in the Peruvian Stock Market to assess the presence of a growing tendency toward a financially fragile situation. Second, at a more aggregate level, by assessing the main drivers of firms' indebtedness growth upon which the Minskyan financial fragility theory relies.

The article is organised as follows. The second section retraces the main analysis of the Minskyan financial fragility theory and considers the main works for the Latin American region, and Peru in particular. The third section focuses on the economic environment of Peru and builds Minskyan indicators to assess the presence of financial fragility at a sectoral level. Section four contains our econometric analysis and finally section five discusses and concludes.

2. Literature review

During the last two decades, Minsky's financial fragility theory and his broader work have become increasingly popular at world level, especially after the 2007–2008 Global Financial Crises (GFC). The Economist (2016), for example, pointed out how 'everyone was turning to his writings as they tried to make sense of the mayhem. Brokers wrote notes to clients about the "Minsky moment" engulfing financial markets. Central banks referred to his theories in their speeches. And he became a posthumous media star, with just about every major outlet giving column space and airtime to his ideas'.

Also in the policy space, many policymakers recognised the relevance of Minsky's work. For example, immediately after the GFC, Janet L. Yellen, President and CEO of the Federal Reserve Bank of San Francisco gave a speech entitled "A Minsky Meltdown: Lessons for Central Bankers" (Federal Reserve Bank of San Francisco, 2009). Similarly, Andy Haldane, Chief Economist and Executive Director of the Bank of England in 2021 considered that if the economy had traversed a change in inflation expectations, it would have paved the way to a change toward a high inflation narrative and ultimately to "a Minsky Moment for monetary policy, a taper tantrum without the taper" with the subsequent increase of interest rates (Bank of England, 2021). More recently, after the continuous FED-interest rate increases, referring to Minsky, the co-founder of Pimco urged the FED to stop raising interest rates further (Gross, 2022) and also JPMorgan warned against Increasing Chances of 'Minsky Moment' (Reinicke, 2023).

However, academic works have been majorly focused on the U.S. and the European economies (for instance Davis et al., 2019; Kregel, 2008; Papadimitriou et al., 2019; Papadimitriou et al., 2020; Nikiforos & Zezza 2017; 2018) and Minsky's warnings have shaken policymakers' concerns majorly in those countries. Instead, relatively little focus has been directed toward the Latin American region. Bibi et al. (2024), Avendaño and Vázquez (2011), Cruz et al. (2006), Kaltenbrunner and Painceira (2015) and Pérez Caldentey et al. (2019) constitute some exceptions in that since they empirically operationalise Minsky's analysis for several Latin American economies. On a broader Latin American and Caribbean (LAC) region context, Pérez Caldentey et al. (2019) offer a very good overview of the indebtedness situation at an institutional level (Government, Central Banks, Financial Corporations, Commercial Banks, Other -private- Financial Corporations, Public Banks, Non-Financial Corporations and Other-public-Non Financial Corporations) between 2000 and 2017 for the different LAC countries. Two major points should be underlined from their study. First, as Fischer (2015) previously also noticed, they highlighted how in the GFC aftermath, Latin American

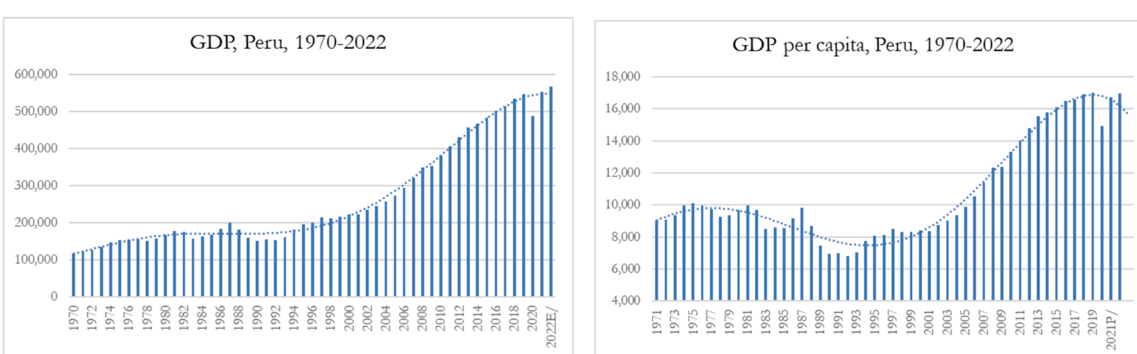

Fig. 1. GDP (left) and GDP per capita (right) of Peru, 1950–2022, constant 2007 prices in Soles.
Source: author elaboration based on INEI database (2022).

Fig. 2. Stock Market Capitalization to GDP for Peru, 2000–2020.
Source: Authors' elaboration based on FRED (2023)

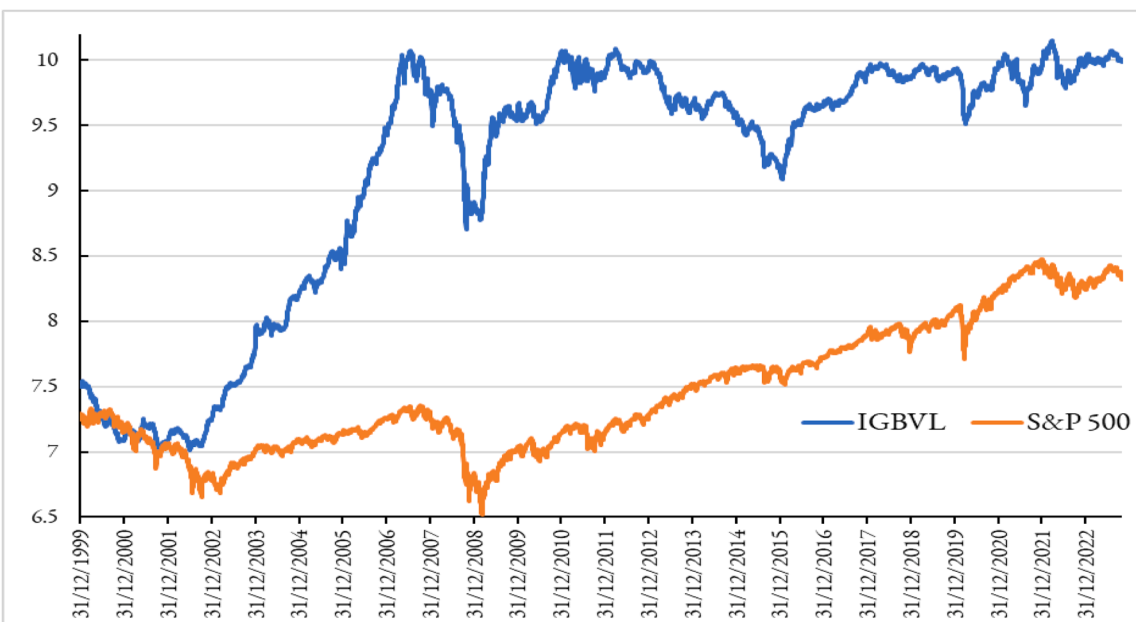

Fig. 3. IGBVL and S&P 500 Index levels (in log scale), 2000–2022.
Source: Data taken from Datastream

economies experienced an increase in liquidity derived from the extraordinary quantitative easing policies in the developed countries. The latter flooded LAC (and other developing) economies with financial resources mainly thanks to their virtual absence of economic contraction, their absence of balance of payment or financial crises and the Latin American favourable commodity prices together with exchange rate dynamics. Second, the increased relevance of financial and more importantly non-financial corporate sectors in many Latin American countries, Peru included. In fact, if in South America government debt declined its relevance as a share of the overall country debt from 72 % to 45 %, the financial corporations increased their level from 8 % to 15 % while non-financial corporations increased from 12 % to 25 % during the same period.

However, to our knowledge, no empirical work has investigated the financial fragility situation of Peru at both aggregate and sectoral levels during the last two decades. The goal of our work is to cover that gap shedding some light on the possible financial fragility dynamics that Peru might have traversed during that period and, at an aggregate level, to highlight the drivers of indebtedness growth upon which the Minskyan financial fragility theory relies on. The study is relevant since it analyses if, beneath the financial stability of Peru portrayed as the Latin American miracle, hidden financial fragility dynamics are instead

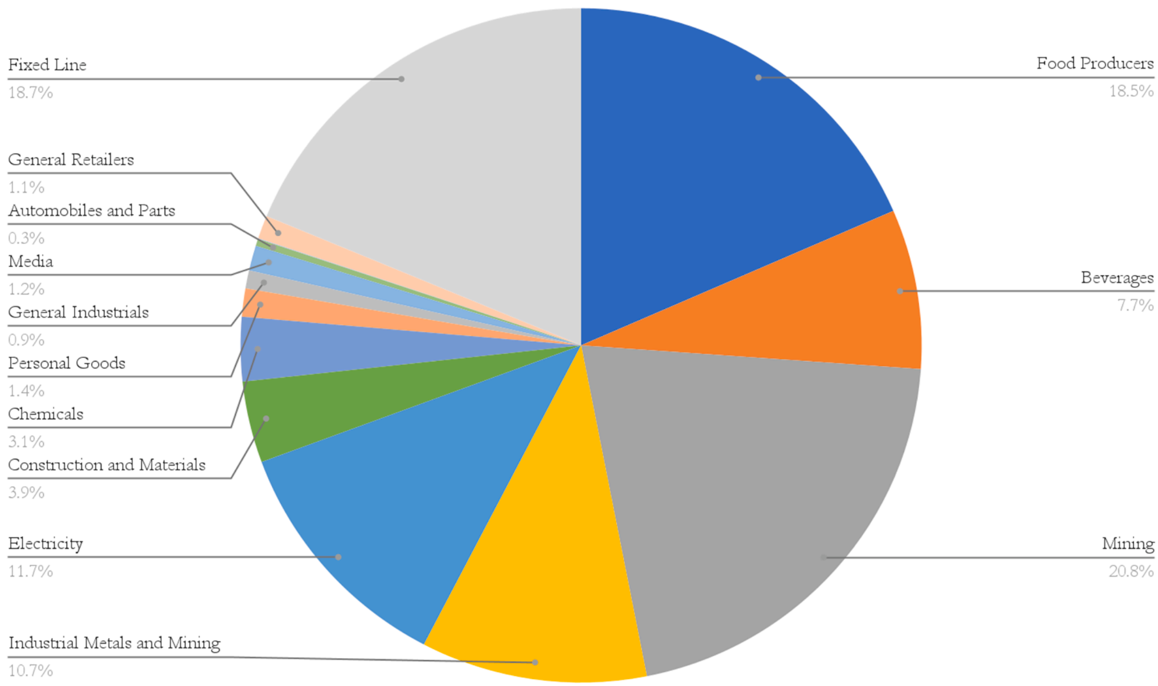

Fig. 4. Average assets by sectors as a proportion of the overall assets, during 2003–2023.
Source: authors' own calculation

building up. In that sense, the cornerstones of our work are Avendaño and Vázquez (2011), Bibi (2024a), Bibi and Valdecantos (2023, 2024) and Bibi et al. (2024).

Bibi, 2024a analyses the overall economic structure of the Peruvian economy and, in retracing similar tendencies with Kazakhstan, it suggests a pattern of financial dependency and subordination for natural resources export-led economies. Bibi and Valdecantos (2023) focus on the relationship between the enclave economic structure of Peru, the supercycle commodity boom and the economic deterioration of the economic conditions after its end that led to increased social claims. Bibi and Valdecantos (2024) study, instead, the financial fragility of Peru at a macroeconomic and institutional level focusing on the financial sector, the non-financial private sector and the public sector. Changing the unit of analysis from individual firms to institutional levels, they build Minskyan indicators of financial fragility. Their work underlines how, while the three sectors altogether were financially stable during the period 2000–2020, the non-financial private sector would have been highly fragile - and in a Ponzi situation during several years - if the commodity prices had remained at their historical level instead of being inflated by the supercycle commodity boom. Finally, based upon Foley (2003), Avendaño and Vázquez (2011) build Minskyan indicators of financial fragility at an individual firms level and Bibi et al. (2024) extend that analysis including sectoral levels for the Mexican NFPS during the period 2005–2021.

Given the special increase in the NFPS indebtedness levels underlined by Pérez Caldentey et al. (2019) for LAC, and the potential financial fragility for that overall sector highlighted by Bibi and Valdecantos (2023, 2024) for Peru in particular, our work focuses the attention on the Peruvian NFPS. Also in order to offer a comparison of financial fragility at a sectoral level inside the Latin American region, here we follow the framework used by Bibi et al. (2024) and Avendaño and Vázquez (2011). Their taxonomy allows us to build indicators to detect the possibility of an increased Minskyan financial fragility phenomenon.

Minsky argued that capitalist economies have a tendency to be endogenously unstable and that famously "stability is destabilizing". The Minskyan Financial Instability Hypothesis (FIH) particularly points to the destabilizing nature of increasing indebtedness that characterizes corporations. In fact, according to Minsky, periods of economic growth are signed by stability and tranquillity where "success breeds the possibility of failure; the absence of serious financial difficulties over a substantial period leads to the development of a euphoric economy" (Minsky, 1986, p. 213). In such a prosperous situation, institutions and regulators tend to relax financial regulations confident that a new stability period has finally arrived.

> As a previous financial crisis recedes in time, it is quite natural for central bankers, government officials, bankers, businessmen, and even economists to believe that a new era has arrived. Cassandra-like warnings that nothing basic has changed, that there is a financial breaking point that will lead to a deep depression, are naturally ignored in these circumstances. Since the doubters do not have fashionable printouts to prove the validity of their views, it is quite proper for established authority to ignore arguments drawn from unconventional theory, history, and institutional analysis. Nevertheless, in a world of uncertainty…the successful functioning of an economy within an initially robust financial structure will lead to a structure that becomes more fragile as time elapses. Endogenous forces make a situation dominated by hedge finance unstable, and endogenous disequilibrating forces will become greater as the weight of speculative and Ponzi finance increases. Minsky (2008, p. 233–238)

In the euphoria and tranquillity period, the confidence of companies increases, and they engage in taking more and more debt to finance even riskier investments. At the same time, thanks to the relaxation of the regulations and driven by their desire to increase profits, commercial banks tend to loan out more credit to firms. That way, corporations' indebtedness level increases and their financial fragility too, transiting from a hedge as the most financially solid position, to a speculative position until arriving at the most financially fragile Ponzi one. To classify firms into the respective Hedge, Speculative and Ponzi positions, we use here the Foley (2003) configuration upon which more recent works rely (Avendaño & Vázquez, 2011; Bibi et al., 2024). We follow such a categorization to offer a comparison of financial fragility at a

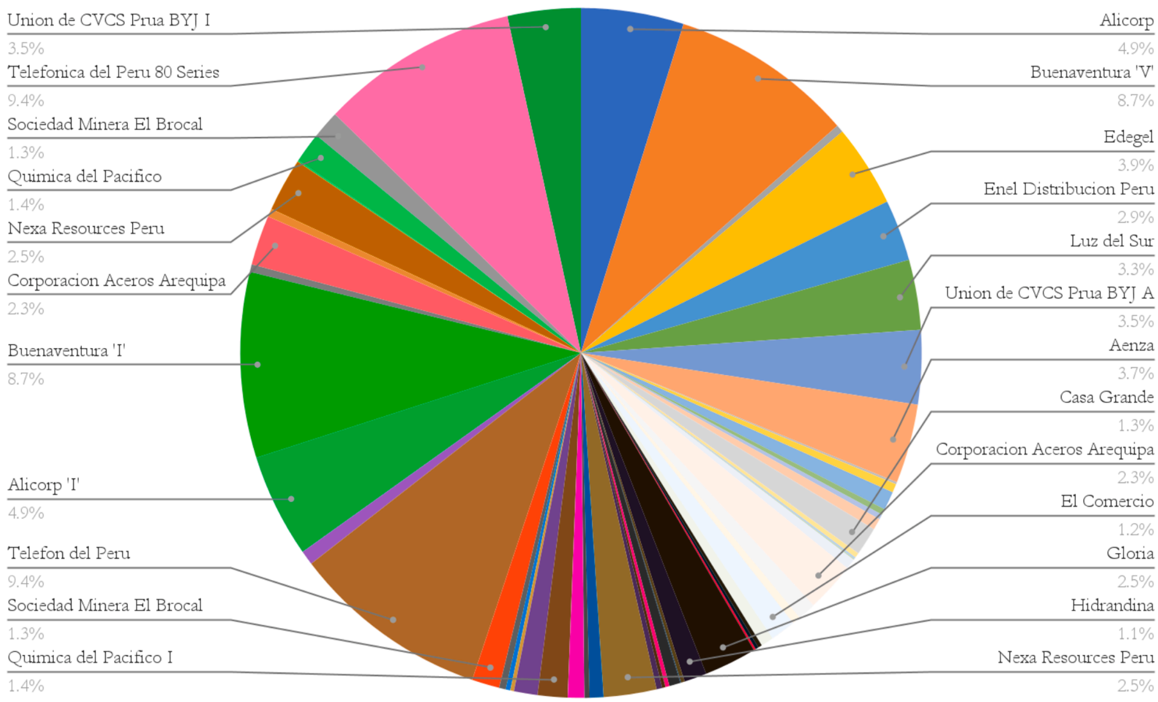

Fig. 5. Average Assets by firm as a proportion of the overall assets, during 2003–2013.
Source: authors' own calculation

sectoral level inside the Latin American region. In fact, Avendaño and Vázquez (2011) analysed the aggregate financial fragility of Mexico between 1990 and 2008 and, more recently, Bibi et al. (2024) studied such a Minskyan financial fragility of that country both at aggregate and sectoral level (2005–2021) capturing both the critical moments of the GFC and the COVID-19 pandemic.

We build indicators to detect the possibility of an increased Minskyan financial fragility phenomenon within the sample of corporations in the Peruvian Stock Market Exchange following Bibi et al. (2024) and Avendaño and Vázquez (2011). The main variables that are used to establish the financial condition of the firms are g, o and ψ. The variable $g = I/A$ represents the corporation's growth of assets over time considering I as investments and A as its assets while $o = O/A$ represents the ROA, namely the corporation's return on assets with O representing the corporation's own gains. Finally, $\psi = \Psi/D$ represents the amount of interest that a corporation needs to pay back over its debts with Ψ representing the amount of interest to be paid and D its indebtedness level.

3. Context: economic environment since 2000

After a tumultuous period signed by hyperinflation (up to over 3000 %) and a strong recession (with peaks higher than 10 % of GDP), since the beginning of the XXI century, Peru seems to have entered a glorious period highlighted and welcomed by many national and international institutions with high GDP growth levels, reduction of poverty and an apparent macroeconomic and financial stability. The overall economy has been majorly driven by its mineral exports which, by themselves, arrived to count >60 % of exports during the 2003–2014 supercycle commodity period (Bibi, 2024c, Bibi and Valdecantos 2023, 2024).

Fig. 4 captures the average assets of the different sectors in the sample studied during the period 2003–2023 revealing indeed the relevance of mining activities (Industrial Metal and Mining sector and Mining one) that, combined, counted for >30 % of the total assets. The other major sectors are the ones related to food production, fixed-line telecommunication and electricity with 19 % and 12 % respectively. Fig. 5 (in the Appendix) gives the same information by individual firms.

We start analysing the overall sample of corporations at an aggregate level by examining the main variables described in Table 1, g, o and ψ. Fig. 6 reveals that o maintains a level structurally and persistently higher than both g and ψ (except for one quarter in 2021), that way leaving no doubt of the condition of companies at the aggregate level. In that context, the firms are able to generate a sufficient amount of earnings higher than both the corporation's growth of assets ($g = I/A$) and the interests needed to pay back over its debts ($\psi = \Psi/D$), therefore portraying the most solid hedge financial situation.

However, Fig. 7 suggests a less straightforward scenario characterized by several periods with a rate of indebtedness growth ($d=\Delta D/D$) much higher than the rate of growth of assets ($g=I/A$).[1] The indebtedness growth rapidly increased from 2003 until the 2007–2008 GFC, then it was reduced and remained lower than the rate of assets growth until the end of the supercycle commodity boom towards 2013, and again decreased slowly after that. Since 2016, two major periods seem to spark attention with respect to the growth rate of companies' indebtedness, namely the pre-and-post-COVID-19 spread in 2020.

The warnings of Fig. 7 suggest looking more deeply at the proportion of firms in a hedge, speculative and Ponzi condition at an aggregate level, following Minsky's hint:

> It can be shown that if hedge financing dominates, then the economy may well be an equilibrium seeking and containing system. In contrast, the greater the weight of speculative and Ponzi finance, the greater the likelihood that the economy is a deviation amplifying system. Minsky, 1992

Figs. 8 and 9 (in the appendix) show the weight of companies in Hedge and Speculative conditions. However, here we focus our attention on the composition of the proportion of the riskiest type of firms, the

[1] The bold lines are the moving averages with 4 periods to help focusing more on the overall dynamic trends rather than the potential cyclical fluctuations described by the dotted lines.

Table 1
Categorization of firms into hedge, speculative and ponzi.

Firm Configuration	Firm Condition	Explanation
Hedge	o > g > ψ or o > ψ > g	A company is in a Hedge situation when its rate of earnings (o=O/A) is higher than both its growth of assets (g=I/A) and the interests it needs to pay back over its indebtedness (ψ = Ψ/D). The Hedge position represents the most financially solid and healthy situation in which a company can be.
Speculative	g > o > ψ	A company is in a Speculative situation when its rate of earnings (o=O/A) is higher than the interests it needs to pay back over its indebtedness level (ψ = Ψ/D) but still lower than its growth of assets (g=I/A). In this condition, despite the company is able to cover the interest payment duties with the generated earnings, those are unable to maintain the speed of growth of the company, namely the growth of assets (g). In this situation, only part of the new investments (I) is covered by the company's generated earnings (O) while the company needs to engage in new loan requests to finance the residual part. The indebtedness activity entails some level of risk and hence the company finds itself in a riskier situation compared to Hedge.
Ponzi	ψ > o	A company is in a Ponzi position when its earnings are insufficient not only to cover the sum of interests needed to be paid plus the new investments, but they are even insufficient to cover the only interests themselves. The rate of earnings (o=O/A) is lower than the interest the company needs to pay back over its indebtedness (ψ = Ψ/D). In this situation, the company needs to engage in loans not only to cover all or part of its principal (I, investment projects) but it needs to run into debt to cover the interests themselves. The Ponzi situation is the most fragile financial condition a company can find itself in and in case that condition is protracted, the company is prone to be unsolvable and collapse.

Source: Adapted from Bibi et al. (2024) and built on Foley (2003) and Avendaño and Vázquez (2011).

Ponzi ones, at a sectoral level.[2] By doing that, Fig. 10 reveals not only the dynamics of Ponzi companies overall (bold line) but also the different sectors that were exposed in different periods to this condition. We argue that it is important to analyse the weight at a sectoral level since the simple analysis at the aggregate one might hide some proper dynamics specific for some sectors. That analysis at sectoral level seems also relevant and in line with Minsky argument about a potential "domino effect" where the financial fragility and eventual collapse of some firms can dangerously spread even to firms much more financially robust, that way contaging other sectors and solid parts of the economy. In Minsky words "*As with a line of dominoes, collapse of the speculative borrowers can then bring down even hedge borrowers, who are unable to find loans despite the apparent soundness of the underlying investments*" (Minsky, 1992). (Fig. 11)

Indeed, in contrast with the apparent stability presented by Figs. 6, 10 shows that despite fluctuating, the average of firms in a Ponzi situation increased from <20 % to almost 40 % during the period under observation. Furthermore, a great divergence in terms of both averages and trend dynamics was present among the different sectors.

First, the mining sector impressively increased the proportion of Ponzi firms from 20 % in 2004 to >60 % during the 2008–2009 period. After declining to more moderate levels below 25 % in 2013, however, the proportion of Ponzi firms in the mining sector rapidly increased again arriving at values of 50 % in 2016 and at peak values of 80 % since 2019. Again, it is worthy to underline that the sharp increase in the proportion of Ponzi in this sector was not triggered by the Covid-19 pandemic since such an increasing path clearly started since the end of the supercycle commodity boom in 2013–2014. A second dynamics that is revealed is the one related to the food sector. Here, while the proportion of Ponzi firms initially fluctuated between 20 % and 30 % until 2010, since then it strongly increased arriving at 50 % in 2013–2014. After that, such a proportion declined toward levels of 20 % at the end of the observed period. Finally, if the sector of industrial metals remained with a relatively stable proportion of Ponzi companies until 2017 (around 20 %), that proportion rapidly increased since then arriving at levels beyond 50 % toward the end of the period observed. The sector of personal goods followed a very similar dynamic. Differently, the sector of electricity did not face relevant surges in the proportion of Ponzi firms and can therefore be considered the sector more financially solid during the whole period. Those dynamics should be considered by policymakers in the moment of taking sectoral tailored policies.

4. Empirical framework and results

In this section we empirically investigate the cornerstone of Minsky's Financial Instability Hypothesis in Peru, namely the growth of indebtedness through time. For that, we use quarterly firm-level data of non-financial companies listed on the Lima Stock Exchange (Bolsa de Valores de Lima). Since this analysis focuses on non-financial companies, we excluded the companies from Banking, Financial Services, Insurance (both Life and Non-life) and Real Estate Investment Services sectors. In total, our sample consists of 58 firms and for each of those companies we have financial data from 2003 Q1 to 2023 Q1 giving us a total of 4968 firm-quarter combinations. The data is taken from Datastream.[3] Table 2 presents summary statistics of the data used in our panel data model (1). Panel A of Table 2 shows the summary while Panel B shows the cross-correlations. In Panel A of Table 2, apart from the usual description of various moments of the distribution of the data, we also report the first-order autocorrelation [AC(1)] and its associated Q-stat which shows that first-order autocorrelation is insignificant. We also report the Levin et al. (2002) t-statistic, LLC(t), in the last row of Panel A which shows that null hypothesis of presence of unit root is rejected. Panel B shows that the cross-correlations among the variables are insignificant.

We estimate model (1) below to investigate the drivers of growth in indebtedness.

$$d_{i,t} = \alpha + \beta_1 g_{it-1} + \beta_2 o_{i,t-1} + \beta_3 \psi_{it-1} + \varphi d_{i,t-1} + \varepsilon_{i,t} \tag{1}$$

where:

$d_{i,t}$-	denotes the growth rate of debt of the i^{th} firm at time *t*,
g_{it-1}-	denotes the growth rate of assets and is calculated as ratio of total Investments to total assets of i^{th} firm at time *t-1*,
$o_{i,t-1}$-	denotes the rate of return and is calculated as the ratio of Income from Operations before tax to total assets of i^{th} firm at time *t-1*,
ψ_{it-1}	denotes rate of interest paid by the i^{th} firm at time *t-1* on its total debt and is calculated as the ratio of Interest Expense to total debt.
$\varepsilon_{i,t}$	is assumed to be a normally distributed IID shocks.
φ	is the autoregressive parameter that measures the dynamics of growth rate of debt.

The empirical investigation using model (1) represents the financial

[2] For readability of the figures, we here represent the major sectors, considered as the ones with companies constituting >10% of the overall sample. In particular, sector 1 is the Food Producers, sector 3 is Mining, sector 4 is Industrial Metals and Mining, sector 5 is Electricity and sector 8 is Personal Goods. Combined together they represent over 70% of the whole sample. Again, for readability, the values shown in Figs. 6, 8 and 9 are moving average values, with 11 periods considered as Bibi et al. (2024) also did. We follow the same moving average periods also to allow for comparisons between the different countries.

[3] Please see table A1 in the Appendix for more details about the data.

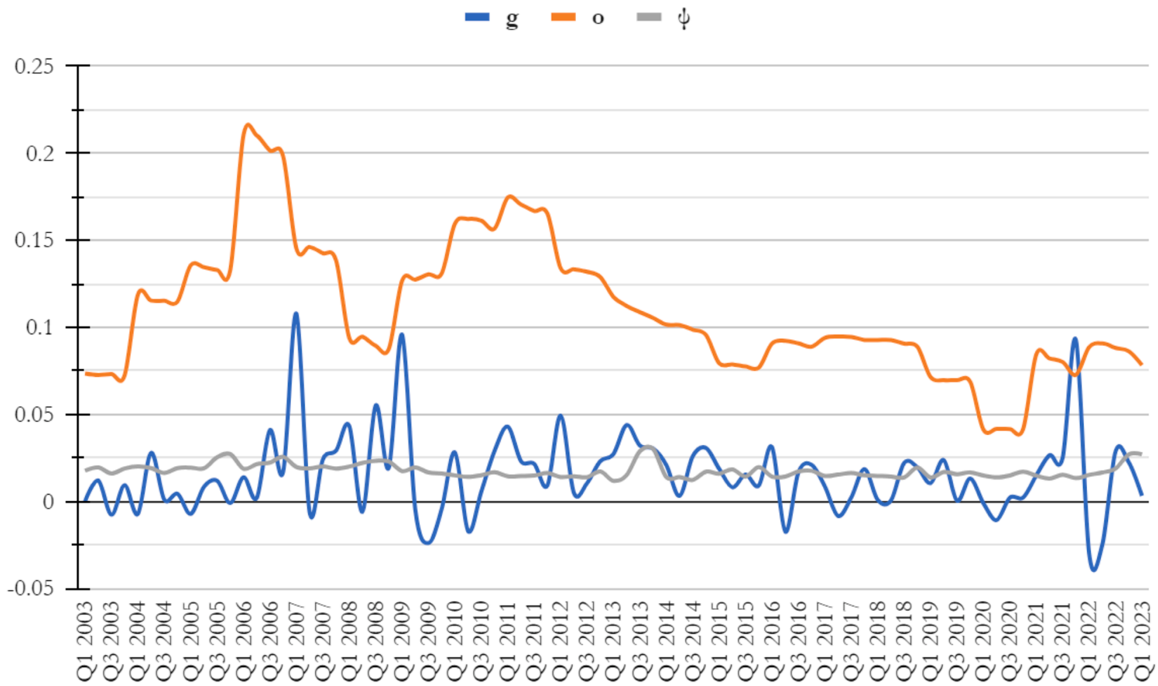

Fig. 6. Profitability, growth and debt burden at an aggregate level, 2003–2023.
Source: authors' own calculation

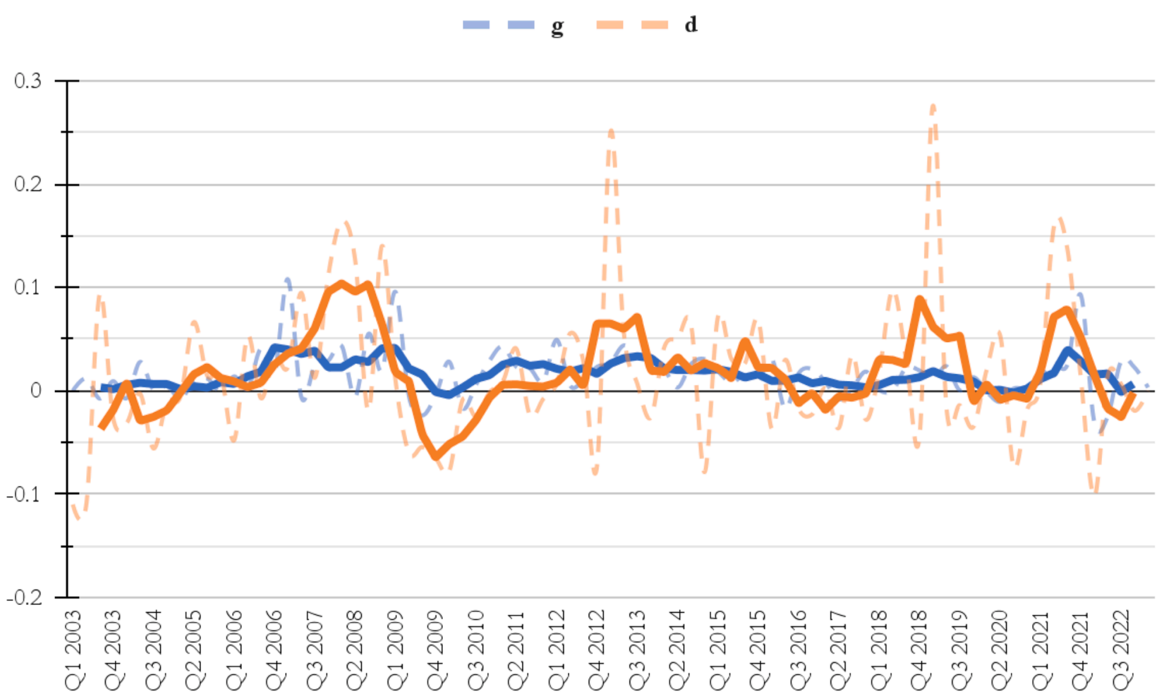

Fig. 7. Growth and indebtedness, aggregate level, 2003–2023.
Source: authors' own calculation

structure approach of Minsky (1982) and relies on the micro interpretation of FIH implying that individual firm-level financial data matter in investigating systemic financial fragility (Toporowski, 2008; Pedrosa, 2019). The main motivation to use dynamic panel data model in our empirical framework is three folds. First, to capture the potential temporal dynamics in the growth rate of debt in the firm-level data. It would not be impractical to assume lagged effects in growth rate of debt levels. Second, since we estimate the Dynamic Panel Model using Arellano-Bond GMM estimator, we address the potential problem of endogeneity. Third, since we use orthogonal deviations in estimating the model parameters, we are accounting for unobserved, time-invariant heterogeneity thus preventing from biasing the results. Thus, our framework is robust to temporal dependencies and unobserved heterogeneity.[4] Furthermore, following Avendaño and Vázquez (2011) and Bibi et al. (2024), the variables used in model (1) are motivated by

[4] We thank the anonymous referee for this.

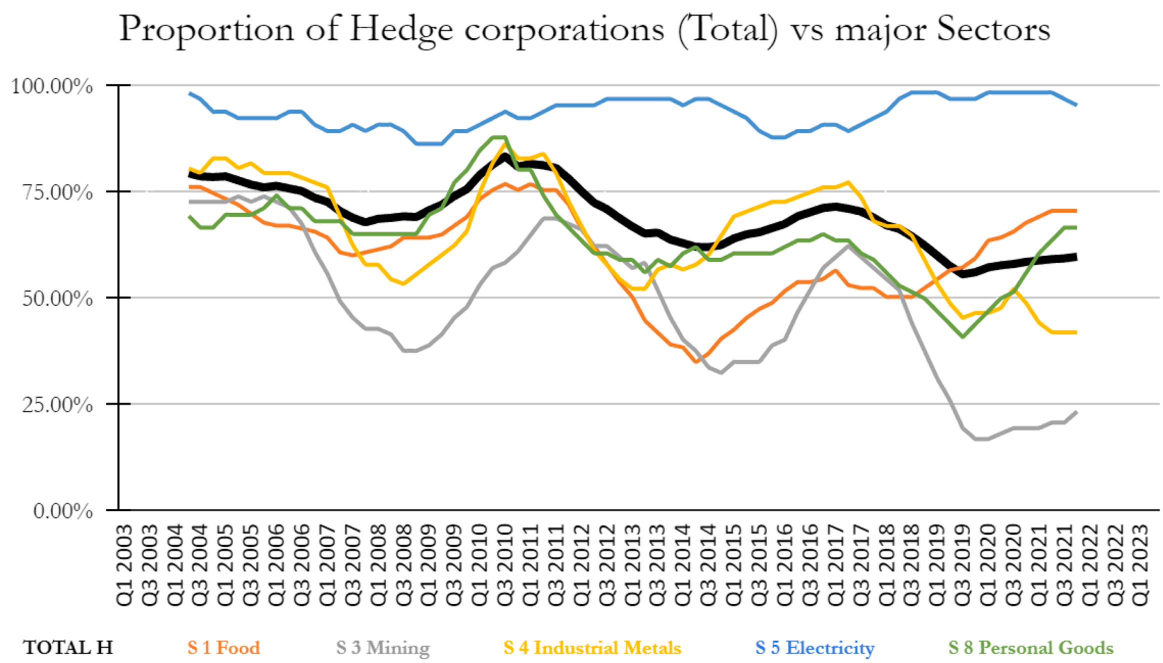

Fig. 8. Proportion of Hedge firms at aggregate and sectoral level, 2003–2023.
Source: authors' own calculation

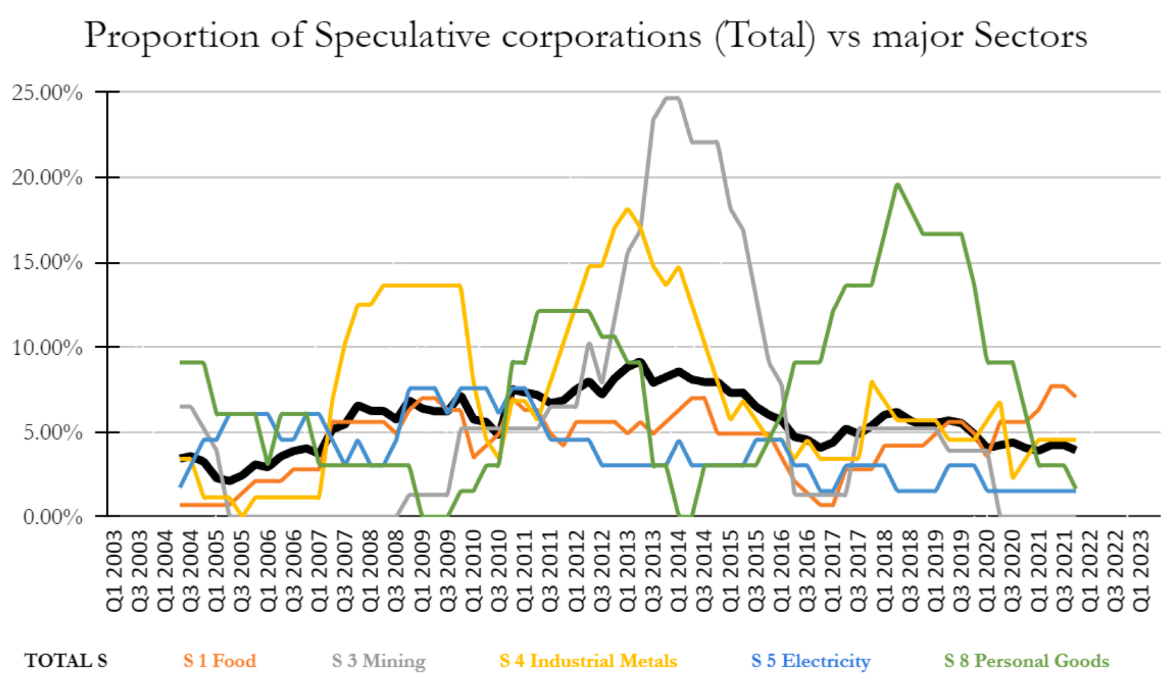

Fig. 9. Proportion of Speculative firms at aggregate and sectoral level, 2003–2023.
Source: authors' own calculation

Foley's (2003) reformulation of the typology in the FIH investigation.

To test the FIH, we focus on two parameters viz. β_1 which measures the sensitivity of the growth rate of debt to the growth rate of assets and β_2 which measures the impact of the rate of return of assets on the growth rate of debt. If the micro interpretation of FIH is true, then β_1 is conjectured to be positive and significant. This would imply that firms will take on more debt to finance the growth rate of assets. Additionally, β_2 is hypothesised to be negative and statistically significant, which would imply that as the firm's rate of return (i.e. income from operations before tax per unit assets) increases, there will be less need for the firm to take on more debt to finance future operations. We estimate model (1) using GMM to address the potential issue of endogeneity. The results of the model (1) for the entire sample period are presented in Table 3. We use second lags of the regressors as instruments. The J-statistic is 53.87 with a p-value of 0.26 and hence we cannot reject the null hypothesis that the instruments are uncorrelated with the errors. As such, the results do not provide evidence against the validity of the instruments.

If we observe Table 3, we can see some interesting results. First, the sensitivity of the growth rate of debt to the growth rate of assets is positive ($\beta_1 = 22.61$) and is statistically significant. This result is in line

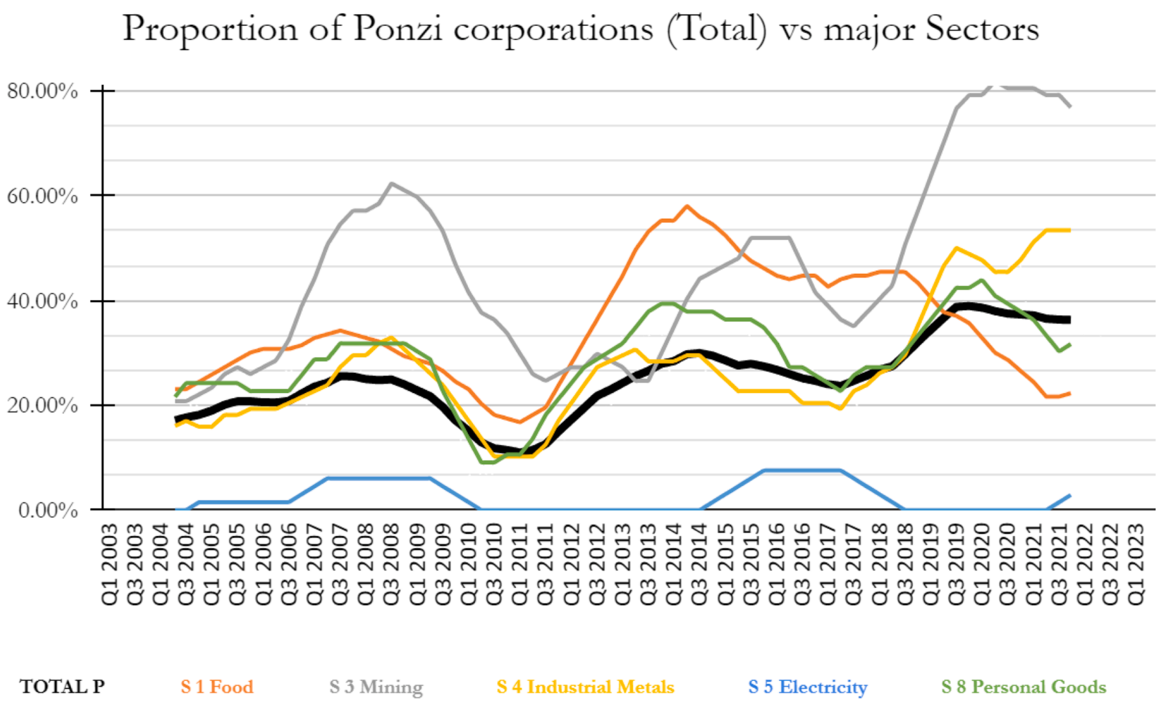

Fig. 10. Proportion of Ponzi firms at aggregate and sectoral level, 2003–2023.
Source: authors' own calculation

Fig. 11. Price of Copper (per ounce) in USD, 1990–2020.
Source: Trading Economics (2024)

with the literature and shows that as the firms grow (i.e. as the growth rate of assets increases), keeping everything else constant, the growth rate of debt also increases implying that they take on more debt. Second, the sensitivity of the growth rate of a firm's debt to the rate of returns from operations is negative and statistically significant (β_2=- 17.07). This implies that as the rate of income from operations before tax of a typical firm increases, keeping everything else constant, the firm's indebtedness decreases, thus highlighting the importance of organic funding/resources. Third, the elasticity of a firm's indebtedness to interest expense is positive and statistically significant (β_3=0.04). This is also in line with the extant literature, and it shows that, as the average interest expense on the existing debt increases, the firm's indebtedness also increases. This could be because firms may issue more debt to finance the cost of the existing debt which may eventually become unsustainable and contribute to higher financial fragility. Fourth, the autoregressive coefficient is negative i.e., (φ =−0.02) and significant.

Table 2
Data summary and cross-correlations.

	d	g	o	ψ
Panel A: Summary Statistics				
Mean	5.85***	0.007	0.12***	3.07***
t-stat	(2.81)	(1.43)	(67.27)	(2.74)
Median	0.000	0.005	0.08	0.017
Std. Dev.	142.52	0.35	0.21	76.67
Skewness	28.56	−62.35	3.12	33.46
Kurtosis	862.87	4134.64	105.4	1144.25
AC(1)	−0.002	0.004	0.15	0.15
Q-stat	0.016	0.065	1.12	1.52
LLC(t)	−26.89***	−22.74***	−1.74**	−8.19***
Panel B: Cross-Correlation				
	d	g	o	ψ
d	1			
t-stat				
g	0.0035	1		
t-stat	(0.23)			
o	−0.014	0.08	1	
t-stat	(−0.34)	(1.23)		
ψ	−0.001	0.0003	0.024	1
t-stat	(−0.12)	(0.026)	(1.63)	

Table 3
Determinants of indebtedness growth rate of companies.

	$o_{i,t-1}$	ψ_{it-1}	$d_{i,t-1}$	J-stat	P(J-stat)
22.61***	−17.07***	0.04***	−0.02***	53.87	[0.26]
(78,647.2)	(−15,523.6)	(869.4)	(−19,668.5)		

Notes: The dynamic panel model is estimated using GMM (orthogonal deviations). Figures in parentheses are the t-statistics adjusted for clustering. The associated standard errors are corrected by the degrees of freedom. Total 4582 firm-quarters observations included. Instruments included are the second lag of the regressors and a constant. Instrument rank is 52. *** denotes significance at 1 % level.

This implies that the growth rate of debt in one period is negatively affected by the growth rate of debt in the previous period. It is important to underline that, even if this is the case, yet, the overall level of debt might still be increasing. In line with other studies on Minsky's FIH in Latin America (Avendaño & Vázquez, 2011; Bibi et al., 2024) and especially considering recent studies on the financial fragility particularly in Peru at more aggregate level (Bibi & Valdecantos 2023; 2024), our work aims to shed light on some particular risk of increased financial fragility at sectoral level. In particular, our analysis shows that although an increase in financial fragility cannot be detected at macroeconomic level, Peru may have faced potential dynamics of financial fragility at sectoral level during different periods (in the mining, food and industrial metal sectors specifically). Because of the relevant weight increase of companies in Ponzi regime in those sectors and because of the "domino effect" pointed out by Minsky, the detection of such sectoral dynamics become relevant for Peru and similarly for many other countries.

5. Discussion and conclusions

In the last two decades Peru has been often portrayed as the Latin American miracle (Mendoza, 2013; Vietor et al. 2015) because of its sound macroeconomic and financial stability. This success has been often ascribed to the mainstream orthodox policies adopted since the 1990s such as increased deregulation and financial liberalisation. Peru saw exceptional economic growth both in terms of GDP and GDP per capita, especially if compared with the neighbouring Latin American countries leading the IMF to stress such a successful story repeatedly (Ross & Peschiera, 2013; IMF, 2008; 2012; 2017; 2023). However, recently, Bibi and Valdecantos (2023, 2024) highlighted the hidden financial fragility of Peru, mainly linked to its non-financial private sector. As such, in this paper, we investigated this idea in more detail.

In particular, our paper contributed to analyse if, beyond a surface of growth and stability, Peru NFPS has been subject to a financial fragility-increasing phenomenon. Minskyan hint of a slow building up financial fragility process in a context of success and tranquillity suggested an analysis both at an overall and sectoral level considering the weight of companies exposed to the more financially fragile Ponzi condition. Our study focuses on a sample of NFPS companies listed in the Peruvian BVL stock market, across different sectors during the 2003–2023 period. Our findings revealed that while at an overall level, a situation of increased financial fragility can not be detected, a more focused analysis at a sectoral level is able to depict interesting and potentially worrying dynamics. In fact, while the profitability rate level (o) is structurally higher than both the companies' growth rate (g) and the amount of interest that they need to pay back over their debts (ψ) (Fig. 6) representing a safe hedge situation at an overall level, different averages and trend dynamics were present among the different sectors. In particular, the most worrying surges and dynamics were the ones that affected the mining sector during the 2003–2009 period, increasing the percentage of companies in Ponzi condition from about 20 % to about 60 % and increasing again arriving to levels higher than 80 % since 2019. It should be underlined that the sharp increase in the proportion of Ponzi in this sector was not triggered by the Covid-19 pandemic since such an increasing path clearly started since the end of the supercycle commodity boom in 2013–2014. The food sector was also affected by a strong increase in the proportion of Ponzi companies that tripled during the period 2010–2014. Finally, a third unveiled dynamic was the increase in Ponzi companies' weight in the industrial metal sector which increased from <20 % up to levels higher than 50 % during the 2010–2022 period.

We finally econometrically tested at an overall level for the drivers of indebtedness growth, the element at the core of the FIH-Financial Instability Hypothesis. In line with the FIH and with other studies on Latin American countries (Avendaño & Vázquez, 2011; Bibi et al., 2024), we found that companies' indebtedness level is positively explained by companies' growth rate, the interest expenses and negatively linked to their profits. This is in line with the FIH based upon companies' attitude to engage in further indebtedness, due to the euphoria attached to the positive growth periods, and to deleverage when increased profits were obtained allowing them to rely on internal funding instead of the external ones.

Our analysis at an overall and sectoral level aims to be relevant not only on a theoretical ground but also for policymakers of Peru welcoming further studies on the topic of financial fragility at both macro and sectoral level to design policies that could consider the different nuanced exposition of risk of the Peruvian sectors.

Our work aims to represent a good baseline for further research and it can be extended in several ways. First, the drivers of the surge of companies in Ponzi-like conditions in each individual sector should be studied in the different periods, especially focusing on the 2007–2008 GFC, the supercycle commodity boom and finally on the COVID-19 pandemic periods. Second, the sample under investigation might be enlarged relying on improved data available in the future. Third, the empirical design of our model is based on Avendaño and Vázquez (2011) which use a similar idea and we wanted to maintain a similar empirical framework to that work for possible countries comparisons within the Latin American context. However, a further research question could study the potential of interaction between debt growth and capital accumulation, together with using other indicators present in the literature that would strengthen or challenge our results. Forth, while the goal of our work is mainly to identify the divergence between the (undetected) financial fragility at overall level and the rising financial fragility at sectoral level in the non financial private sector, further research might investigate the origin for such disconnection. Finally, our work might well serve as a cornerstone for other neighbouring Latin

American countries in the face of the economic and social turmoil period as well as the changes toward more aggressive neoliberal policies that recently elected governments are likely to push forward.

Funding

No sponsor or funding was received.

CRediT authorship contribution statement

Samuele Bibi: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Pankaj Avinash Chandorkar:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

No potential conflict of interest was reported by the authors.

All the data used are publicly available and downloadable from DataStream, INEI and FRED websites.

Appendix

Table A1
Data description.

Sr No.	Data	Datastream Code	Interpretation/Explanation
1	Total Assets	WC02999	Total Assets represent the sum of total current assets, long term receivables, investment in unconsolidated subsidiaries, other investments, net property plant and equipment and other assets.
2	Total Liabilities	WC03351	Total Liabilities represent all short- and long-term obligations expected to be satisfied by the company. It includes but is not restricted to Current Liabilities, Long Term Debt, Provision for Risk and Charges (non-U.S. corporations), Deferred taxes, Deferred income, other liabilities, Deferred tax liability in untaxed reserves (non-U.S. companies) and Pension/ Post retirement benefits. It excludes Minority Interests, Preferred stock equity, Common stock equity and non-equity reserves
3	Earnings before interest and taxes (EBIT)	WC18191	EBIT represent the earnings of a company before interest expense and income taxes. It is calculated by taking the pre-tax income and adding back interest expense on debt and subtracting interest capitalized.
4	Interest Expense	WC01251	Interest Expense represents the service charge for the use of capital before the reduction for interest capitalized. It includes but is not restricted to interest expense on short term debt, interest expense on long term debt and capitalized lease obligations and amortization expense associated with the issuance of debt.
5	Total Debt	WC03255	Total Debt represents all interest bearing and capitalized lease obligations. It is the sum of long- and short-term debt.

Note: This data is used in constructing the variables described in Table 1 for firm categorisation and for the empirical framework described in Section 4.

Data availability

Data will be made available on request.

References

Avendaño Vargas, B.L., Vázquez Muñoz, J.A., 2011. Inestabilidad financiera y política monetaria en México, 1990-2008. Investig. Econ. 70 (275), 63–92.

Bank of England, 2021. Thirty years of hurt, never stopped me dreaming. Bank of England. https://www.bankofengland.co.uk/speech/2021/june/andy-haldane-speech-at-the-institute-for-government-on-the-changes-in-monetary-policy.

Bibi, S., Valdecantos, S., 2023. The price (and Costs) of macroeconomic stability in Peru: some lessons on the implications of FDI-driven growth. Dev. Change. https://doi.org/10.1111/dech.12793.

Bibi, S., & Valdecantos, S. (2024). Minsky meeting Prebisch: The macroeconomic challenges of Peru in the 21st century. *Available at SSRN 4599012*. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4599012.

Bibi, S., Villanueva, L., & Bucio, C. (2024). Mexico: A Minskyian Case of Financial Fragility Shaken by Covid-19. *Available at SSRN 4396367*. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4396367.

Bibi, S., (2024a). Oil revenues, FDI and Balance of Payment dynamics: the case of Kazakhstan between the Supercycle Commodity boom and Financial Subordination. Resour. Policy 90, 104789. https://doi.org/10.1016/j.resourpol.2024.104789.

Bibi S. (2024b). Patterns of subordination beyond geographical boundaries: the common path of Peru and Kazakhstan between natural resources, foreign ownership and financial inflows. *Available at SSRN 4598184*. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4598184.

Bibi, S., 2024c. Prebisch and the terms of trade. Resources Policy 90, 104813.

Cruz, M., Amann, E., Walters, B., 2006. Expectations, the business cycle and the Mexican peso crisis. Cambridge J. Econ. 30 (5), 701–722.

Davis, L.E., Paulo, J., De Souza, A., Hernandez, G., 2019. An empirical analysis of Minsky regimes in the US economy. Cambridge J. Econ. 43, 541–584. https://doi.org/10.1093/cje/bey061.

Federal Reserve Bank of San Francisco, 2009. A Minsky Meltdown: lessons for Central Bankers. Federal Reserve Bank of San Francisco. https://www.frbsf.org/research-and-insights/publications/economic-letter/2009/05/minsky-central-bank-asset-price-bubbles/.

Fischer, A.M., 2015. The end of peripheries? On the enduring relevance of structuralism for understanding contemporary global development. Dev. Change 46 (4), 700–732. https://doi.org/10.1111/dech.12180.

Foley, D., 2003. Financial fragility in developing economies. Dev. Econ. Struct. Macroecon. 157–168.

FRED, 2023. Stock Market Capitalization to GDP for Peru. https://fred.stlouisfed.org/series/DDDM01PEA156NWDB.

Gross, B., 2022. The Fed needs to stop raising rates. The Financial Times. https://www.ft.com/content/c3cb8378-cac0-4bb1-be7c-170ca002f9fb.

IMF, 2008. Regional Economic Outlook, Western Hemisphere Grappling with the Global Financial Crisis. International Monetary Fund, Washington, DC.

IMF, 2012. Regional Economic Update: Latin America and the Caribbean. International Monetary Fund, Washington, DC.

IMF, 2017. Regional Economic Outlook, Western Hemisphere: Tale of Two Adjustments. International Monetary Fund, Washington, DC.

IMF, 2020. IMF Country Report. International Monetary Fund, Washington, DC. www.imf.org/en/Publications/CR/Issues/2020/01/13/Peru-2019-Article-IV-Consultation-Press-Release-Staff-Report-Staff-Statement-and-Statement-48942.

IMF. (2023). IMF executive board completes review of Peru's performance under the flexible credit line arrangement. https://www.imf.org/en/News/Articles/2023/05/18/pr23160-peru-imf-executive-board-completes-review-of-performance-under-the-flexible-credit-line.

INEI, 2020. Panorama de la Economía Peruana. INEI-Instituto Nacional de Estadística e Informática. https://www.inei.gob.pe/media/MenuRecursivo/publicaciones_digitales/Est/Lib1726/Libro.pdf.

Kaltenbrunner, A., Painceira, J.P., 2015. Developing countries' changing nature of financial integration and new forms of external vulnerability: the Brazilian experience. Cambridge J. Econ. 39 (5), 1281–1306.

Kregel, J., 2004. External financing for development and international financial instability. UNCTAD 32.

Kregel, J., 2008. Using Minsky's cushions of safety to analyze the crisis in the U.S. subprime mortgage market. Int. J. Political Econ. 37 (1), 3–23.

Levin, A., Lin, C.F., Chu, C., 2002. Unit root tests in panel data: asymptotic and finite-sample properties. J. Econ. 108 (1), 1–24. https://doi.org/10.1016/S0304-4076(01)00098-7.

Mendoza, W., 2013. The Peruvian miracle: good luck or good policies? | Economia. Econ. J. 36 (72). https://revistas.pucp.edu.pe/index.php/economia/article/view/7699.

Minsky, H.P., Kaufman, H., 2008. Stabilizing an Unstable Economy, 1. McGraw-Hill, New York.

Minsky, H.P, 1982. The financial-instability hypothesis: capitalist processes and the behavior of the economy. Hyman P. Minsky Archive. 282. https://digitalcommons.bard.edu/hm_archive/282.

Minsky, H.P., 1986. Stabilizing an Unstable Economy. Yale University Press, New Haven, CT.

Minsky H.P. (1992). The financial instability hypothesis. Levy Economics Institute-Working Paper, 74. http://www.levyinstitute.org.

Nikiforos, M., Zezza, G., 2017. The Trump Effect: Is This Time Different? Levy Economics Institute of Bard College, Annandale-on-Hudson, NY. Strategic Analysis.

Nikiforos, M., Zezza, G., 2018. 'America First,' Fiscal Policy, and Financial Stability. Levy Economics Institute of Bard College, Annandale-on-Hudson, NY. Strategic Analysis.

Pérez Caldentey, E., Favreau Negront, N., Méndez Lobos, L., 2019. Corporate debt in Latin America and its macroeconomic implications. J. Post. Keynes. Econ. 42 (3), 335–362.

Papadimitriou, D.B., Nikiforos, M., Zezza, G., 2019. Can Redistribution Help Build a More Stable economy? Strategic Analysis. Levy Economics Institute of Bard College.

Papadimitriou, D.B., Nikiforos, M., Zezza, G., 2020. Prospects and Challenges for the US Economy: 2020 and Beyond. Strategic Analysis. Levy Economics Institute of Bard College, Annandale-on-Hudson, NY.

Pedrosa, Í., 2019. Firms' leverage ratio and the Financial Instability Hypothesis: an empirical investigation for the US economy (1970–2014). Cambridge J. Econ. 43 (6), 1499–1523.

Reinicke, C., 2023. 'Minsky Moment' Chances Increasing, JPMorgan's Marko Kolanovic Says. Bloomberg. https://www.bloomberg.com/news/articles/2023-03-20/jpmorgan-s-kolanovic-sees-increasing-chances-of-minsky-moment?embedded-checkout=true.

Ross, K., Peschiera, J.A., 2015. Explaining the Peruvian Growth Miracle. In: Werner, A. M., Santos, A. (Eds.), Peru: Staying the Course of Economic Success. International Monetary Fund, Washington, DC, pp. 37–50.

The Economist. (2016) Minsky's moment, available at http://www.economist.com/news/economics-brief/21702740-second-article-our-series-seminal-economic-ideas-looks-hyman-minskys.

Toporowski, J., 2008. Minsky's 'induced investment and business cycles. Cambridge J. Econ. 32 (5), 725–737.

U.S. Department of Commerce, 2022a. Peru - Market Challenges. U.S. Department of Commerce. https://www.trade.gov/country-commercial-guides/peru-market-challenges.

U.S. Department of Commerce, 2022b. Peru - Market Overview. U.S. Department of Commerce. https://www.trade.gov/knowledge-product/exporting-peru-market-overview.

Vietor, R.H.K., D'Alessio, F.A., Pino, R.M., 2015. Peru: Economic Miracle or Just a Mirage? Harvard Business School Case 716–728.